Impact of COVID-19 on Standardized Test Scores in Virginia (2015 – 2024)

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ABSTRACT

Standardized testing has gained increased importance in measuring school quality since the passage of the No Child Left Behind Act in 2001. This study examines the impacts of funding sources, budgetary allowances, and the impacts of COVID-19 on middle grade student test performance between 2015 to 2024.

INTRODUCTION & LITERATURE REVIEW

Virginia measures student achievement with annual Standards of Learning (SOL) exams. In 2020 the COVID-19 pandemic caused extended school closures and unprecedented disruptions to student learning. When students returned to school, their performance on the standardized tests were significantly below the pre-pandemic averages. The Virginia Joint Legislative Audit and Review Commission found that the school closures caused a rise in chronic absenteeism, increased mental health challenges, and a marked decrease in student achievement [1]. Some studies indicate the persistent learning loss may be equivalent to 0.5 years of lost instruction [2]

METHODOLOGY

Standardized testing data was downloaded from the Virginia Department of Education website [3]. This data was cleaned in Python, inspected in AWS's Glue DataBrew service, queried in a MySQL database and analyzed with R. Graphics were created with the ggplot package in R and the GeoPandas package in Python.

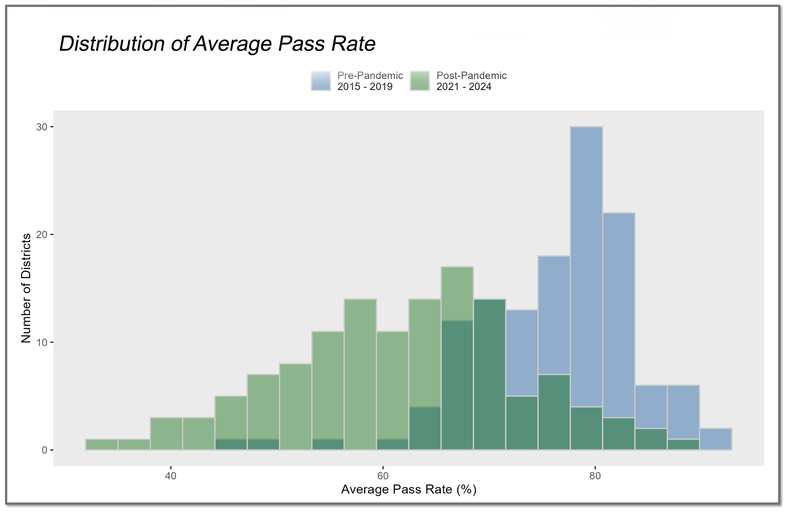

Figure 3 Distribution of annual average pass rate by district before and after pandemic-related school closures

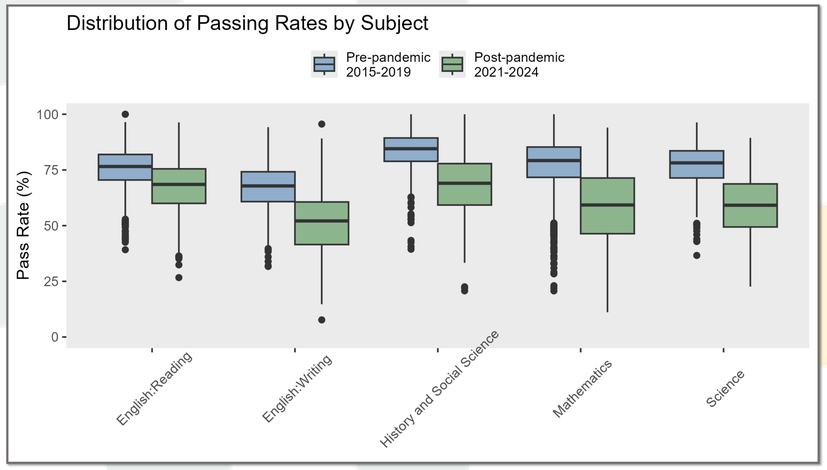

Figure 2: Distribution of average pass rate before and after COVID-19 pandemic

RESULTS

There was a significant decline in SOL scores beginning in 2021. The largest drops were seen in the Mathematics SOL, where the average pass rate dropped from 77.3% before the pandemic to 46.7% in 2021. While the math pass rate has improved over time, the 2024 average pass rate for the Mathematics SOL was still below the pre-pandemic levels at 64.4%. While learning losses were widespread across the state, the school districts in southwestern Virgnia typically declined less and recovered more quickly than the rest of the state (Figure 3).

CONCLUSIONS

Standardized declined sharply, but not uniformly, across the state after the COVID-19 pandemic. More study is needed to establish why some districts were hard-hit with learning loss that continues five years after the school closures, while other districts experienced minor learning loss and quickly recovered to their pre-pandemic averages.

How did COVID-19 impact standardized test scores for Virginia's middle grade students?

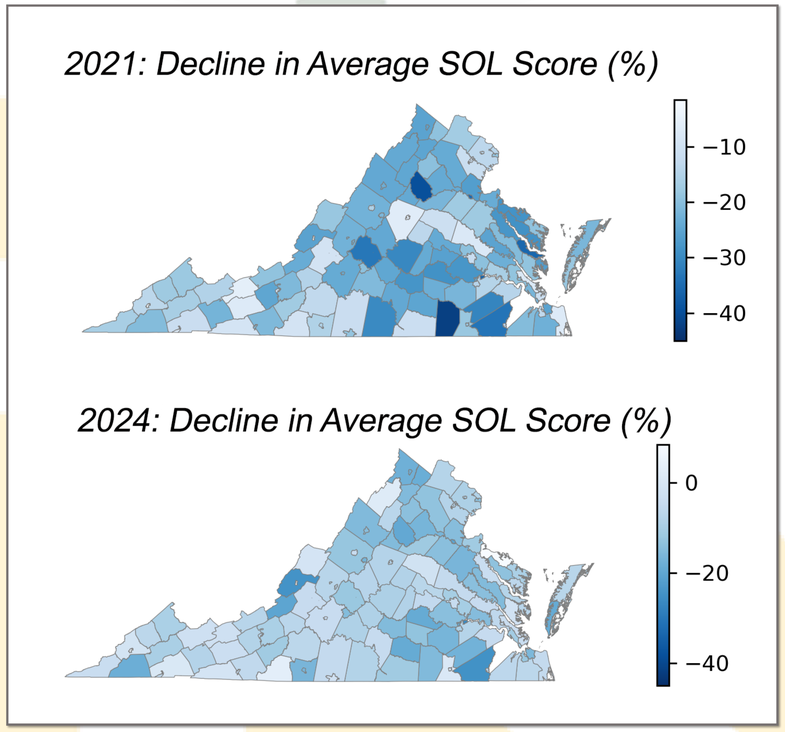

Figure 3: Decline in average pass rate after COVID-19

ACKNOWLEDGEMENTS

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[1] Joint Legislative Audit and Review Commission. "JLARC | Pandemic impact on K-12 Public Education.". https://jlarc.virginia.gov/landing-2022-pandemic-impact-on-k-12-education.asp

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[3] Virginia Department of Education. (2025). "SOL Test Results". https://www.doe.virginia.gov/data-policy-funding/data-reports/statistics-reports/sol-test-pass-rates-other-results (Accessed Aug 25, 2025